

THE UNIVERSITY OF NEW SOUTH WALES

COMP3153/9153 Algorithmic Verification

Final Take-Home Exam

Term 2, 2023

Time Allowed: 24 Hours

Total Number of Questions: 8

Total Number of Pages: 6 (including this title page)

Total Marks Available: 100

Answer **all** questions. The questions **are not** of equal value.

Produce a PDF file, either via L^AT_EX or otherwise, with your prose answers.

Diagrams may be hand-drawn and can be included in the pdf or zipped together with the pdf.

Submit with `give cs3153 exam exam.pdf/exam.zip` or with the give web interface.

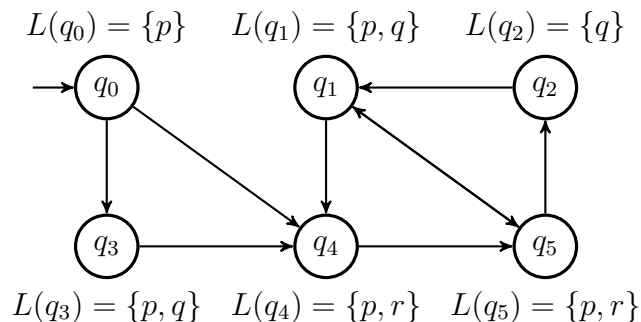
Include the following statement in your PDF file:

I declare that all of the work submitted for this exam is my own work, completed without assistance from anyone else.

You must adhere to the UNSW student conduct requirements listed at <https://student.unsw.edu.au/conduct>.

Question 1 (10 marks)

Consider the following automaton C :



For each of the following CTL* formulae, determine whether C satisfies it or not. Justify your answers.

- (a) $(\mathbf{F} q) \rightarrow (\mathbf{G} p)$
- (b) $(\mathbf{F} \mathbf{G} p) \rightarrow (\mathbf{G} \mathbf{F} q)$
- (c) $\mathbf{E}(\mathbf{G}(\mathbf{F} \neg p))$
- (d) $\mathbf{AG}(p \rightarrow \mathbf{AX} q)$
- (e) $\mathbf{G}((\mathbf{X} q) \vee (p \mathbf{U} r))$

Question 2 (10 marks)

Consider the automaton C from the previous question. The labelling function defines an abstraction, A , of C where the states of A are given by the values of the atomic propositions p , q , and r . For example, because $L(q_1) = L(q_3)$, states q_1 and q_3 would map to the same abstract state.

- (a) Draw A .
- (b) Are A and C bisimilar? Justify your answer.
- (c) Give an ACTL formula, φ , that holds in one of A or C and does not hold in the other.
- (d) Refine the abstraction of A based on φ . That is, find a model A' such that $A \sqsubseteq A' \sqsubseteq C$ and $A' \models \varphi$ if and only if $C \models \varphi$. Try to find a model $A' \neq C$, or explain why this is not possible.

Question 3 (10 marks)

For each of the following pairs of LTL or CTL formulas, prove or disprove whether they are equivalent. That is, prove or disprove for all models $\mathcal{M}$, $\mathcal{M} \models \varphi_1$ if and only if $\mathcal{M} \models \varphi_2$

- (a) $\varphi_1 = \mathbf{G}(\mathbf{F}(p \mathbf{U} q))$ and $\varphi_2 = \mathbf{G}(\mathbf{F} q)$
- (b) $\varphi_1 = (\mathbf{F} p) \mathbf{U} q$ and $\varphi_2 = (\mathbf{F} p) \wedge (\mathbf{F} q)$
- (c) $\varphi_1 = \mathbf{A} \mathbf{X} \mathbf{E}(p \mathbf{U} q)$ and $\varphi_2 = \mathbf{E}(p \mathbf{U} (\mathbf{A} \mathbf{X} q)) \vee q$
- (d) $\varphi_1 = \mathbf{X}(p \mathbf{U} (\mathbf{G} q))$ and $\varphi_2 = \mathbf{G}((\mathbf{X} p) \mathbf{U} (\mathbf{G} q))$
- (e) $\varphi_1 = (\mathbf{X} p) \mathbf{U} (\mathbf{X}(p \mathbf{U} q))$ and $\varphi_2 = (\mathbf{X}(p \mathbf{U} q)) \mathbf{U} (\mathbf{X} q)$

Question 4 (15 marks)

We extend LTL by a **Before** operator. Its semantics are given as

$$\rho \models \varphi \mathbf{Before} \psi \iff (\forall k \geq 0. (\rho[k] \models \psi) \implies (\exists j. 0 \leq j \leq k \wedge \rho[j] \models \varphi)) .$$

- (a) Briefly describe in your own words the meaning of **Before**.
- (b) Prove that the **Before** operator does not extend the expressiveness of LTL. That is, give an LTL formula that is equivalent to $\varphi \mathbf{Before} \psi$ and argue their equivalence.
- (c) Prove the following:
 - i. $((p \mathbf{Before} q) \wedge q)$ implies p ;
 - ii. $p \mathbf{Before} (q \vee r)$ is equivalent to $((p \mathbf{Before} q) \wedge (p \mathbf{Before} r))$.
- (d) Give a Büchi automaton that accepts the words described by the formula $p \mathbf{Before} q$
- (e) How could you extend the LTL model checking algorithm to handle the **Before** operator directly?
- (f) If we add the **Before** operator to the path formulae of CTL in the obvious way, explain how to extend the CTL model checking algorithm to handle $\mathbf{E}(\cdot \mathbf{Before} \cdot)$ subformulae directly.

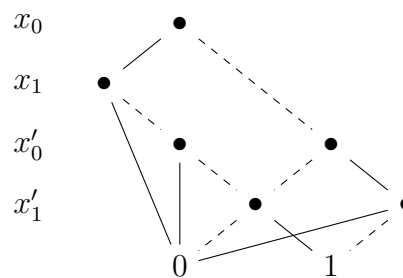
Question 5 (15 marks)

(a) Give OBDDs for the following boolean functions:

i. $f(x, y, z) = (x \wedge y \wedge z) \vee (\neg x \wedge \neg y \wedge \neg z)$

ii. $f(x, y, z) = (x \vee y) \wedge (y \vee z) \wedge (z \vee x)$

(b) Consider the following OBDD describing the boolean function $T(x_0, x_1, x'_0, x'_1)$:



Here a solid line indicates the Boolean value **true**, and a dashed line the value **false**.

$T(x_0, x_1, x'_0, x'_1)$ defines a transition relation of a four state automaton where the states are given by pairs of boolean values (i.e. $Q = \{00, 01, 10, 11\}$). with a transition between $\mathbf{q}$ and $\mathbf{q}'$ if and only if $T(\mathbf{q}, \mathbf{q}') = \text{true}$.

Draw this automaton.

- (c) How can you use OBDDs to solve the SAT problem? That is, given an OBDD for boolean formula φ , D_φ , how can you find an assignment of **true/false** to the variables of φ which makes φ true?
- (d) Prove or disprove: If D_1 and D_2 are OBDDs on n variables with $O(n)$ nodes, then $D_1 \otimes D_2$ has $O(n)$ nodes.

Question 6 (10 marks)

Consider the following decorated pseudo-code program written in a simple imperative programming language:

```
[a:=x+b]1;
[y:=a*b]2;
while [y>a+b]3 do (
    [x:=a+1]4;
    [y:=a+b]5;
    if [(a+1 > a+b)]6:
        [a:=x+1]7;
    else:
        [x:=a+1]8;
);
```

- (a) Give a control flow graph for this program. The code decorations given by $[..]^\#$ indicate the eight critical states to include in the graph.
- (b) Give the table of **gen** and **kill** sets for an **live variable analysis** of the program.
- (c) Compute the result of the live variable analysis, using *backwards may* analysis.

Question 7 (10 marks)

Consider the LTL formula:

$$\varphi = \mathbf{G} (p \mathbf{U} q)$$

Let $\Sigma = 2^{\{p,q\}} = \{\emptyset, \{p\}, \{q\}, \{p, q\}\}$

- (a) Give a Büchi automaton that accepts the words (in Σ^ω) described by φ . Justify your construction.
- (b) Give a Büchi automaton that accepts the words (in Σ^ω) described by $\neg\varphi$. Justify your construction.
- (c) Give a safety property P_s and a liveness property P_e such that the set of behaviours described by φ is equal to $P_s \cap P_e$.

Question 8 (20 marks)

Consider the network of timed automata depicted in Figure 1.

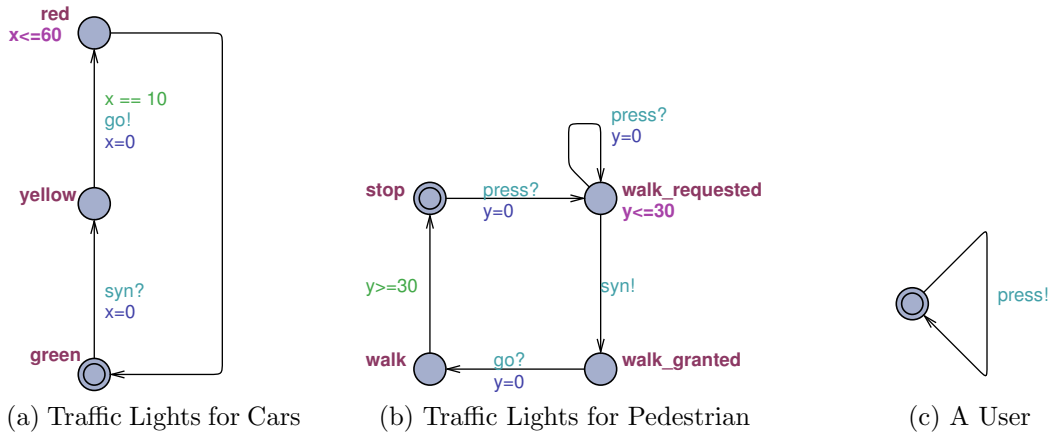


Figure 1: System of Traffic Lights

The automata describe a system of traffic lights. The syntax is as follows: green statements, e.g. $x==10$, are guards; blue statements, e.g. $y=0$, are assignments; purple expressions, e.g. $x<=60$, are invariants of locations; purple names, e.g. **red**, are names of locations; communication channels are light blue, e.g. **press?**. The double-circled locations are the initial states.

There are two different clocks x and y ; moreover the automata make use of the three synchronization channels **press**, **syn** and **go**. Every time unit corresponds to a second.

- (a) Use English to describe each automaton separately.

For the remaining questions we consider the full (synchronised) system.

If you are asked to modify the system, only the parts that are changed need to be shown.

- (b) As discussed in the lecture, a location of a timed automaton is a deadlock state if there are no (enabled) outgoing action-transitions. Does the system contain a deadlock? If yes, describe the situation when a deadlock can occur and modify the system such that no deadlocks can occur.
- (c) Formalise the following properties in CTL, TCTL, or TCTL_c. You can use the names of locations if you like to characterise that the system is in a particular location; you can also use properties on clocks such as $x > 42$.
 - i. At no time pedestrians walk (across the street) while cars have green lights.
 - ii. The yellow light is never turned on for more than 10 seconds.
 - iii. After pressing the button a pedestrian is able to walk across the street within 40 seconds.
- (d) Which of the properties in (c) hold for the deadlock-free system; which do not hold. Justify your answers.
- (e) Modify the system in a way that all properties of (c) are valid.
- (f) Extend the model by a “pedestrian cheat”: If a user presses the button three times within 3 seconds the system switches to a state where cars have to stop (state **red**) and pedestrians can walk.

Don't forget to submit your work and to include the statement listed on the front page.

END OF EXAM